

Functional Dependencies, Normal Forms, and Normalization

Exercise 1

Consider a database of a transport company having the following relation :

Schedule(IdD, Line, Day)

Schedule describes the assignments of drivers (given by their identifier **IdD**) to bus lines (**Line**) for the days (**Day**). For example, in an instance of this schema, the presence of a tuple (05436, 88, 13/01/2020) means that the driver with identifier 05436 is assigned to bus line 88 on the 13/01/2020.

Express the following properties as functional dependencies, when possible. To help you understand the property, give an instance of **Schedule** that does not satisfy it.

1. A driver cannot work on two different bus lines on the same day.
2. A driver is assigned to the same lines every day of the month he works.
3. A driver is assigned to only one bus line.
4. A bus line is always driven by the same drivers.
5. A bus line is driven by a single driver.
6. A driver cannot be assigned to the same bus line on two different days.

Exercise 2

For each of the following relations and the set of functional dependencies associated with it, (a) use Armstrong's axioms to derive all non-trivial FDs from $\mathcal{F}_i$, (b) compute closures of all subsets of attributes, and (c) give the keys.

1. $S(A, B, C, D)$, $\mathcal{F}_1 = \{A \rightarrow B, B \rightarrow C, B \rightarrow D\}$.
2. $T(A, B, C, D)$, $\mathcal{F}_2 = \{AB \rightarrow C, BC \rightarrow D, CD \rightarrow A, AD \rightarrow B\}$.
3. $U(A, B, C, D)$, $\mathcal{F}_3 = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$.

Exercise 3

Are the following relations in 3NF or BCNF? In the case of a negative answer, for questions 1 and 2, provide an instance containing redundancy related to non-conformance to the normal form.

1. $Supplier(Name, Address, Item, Price)$ with $Dom(Name) = Dom(Address) = string$, $Dom(Item) = Dom(Price) = integer$, and the set of associated functional dependencies $\mathcal{F} = \{Name \rightarrow Address\}$.
2. $Wine(Cru, Type, Client, Discount)$ with $Dom(Cru) = \{Bordeau, Riesling\}$, $Dom(Type) = \{Red, Rose, White\}$, $Dom(Client) = string$, and $Dom(Discount) = integer$, and the set of associated functional dependencies $\mathcal{F} = \{Cru \rightarrow Type\}$.
3. $Car(Make, Model, Power, Color)$ with $Dom(Make) = \{Renault, Peugeot\}$, $Dom(Model) = string$, $Dom(Power) = integer$, and $Dom(Color) = \{Red, Black, Blue\}$, and the set of associated functional dependencies $\mathcal{F} = \{Make, Model \rightarrow Power, Power \rightarrow Color\}$.
4. $Wine(Cru, Type, Client, Discount)$ with $Dom(Cru) = \{Bordeau, Riesling\}$, $Dom(Type) = \{Red, Rose, White\}$, $Dom(Client) = string$, and $Dom(Discount) = integer$, and the set of associated functional dependencies $\mathcal{F} = \{Cru, Type, Client \rightarrow Discount, Discount \rightarrow Cru\}$.

Exercise 4

Consider the attributes E (Employee), R (Manager), D (Department), P (Project) and the schema consisting of the relation $R(E, R, D, P)$ with the set of associated dependencies $\mathcal{F} = \{E \rightarrow D, D \rightarrow R, E \rightarrow R\}$. A tuple (e, p, d, r) has the specification *the employee e works on project p in department d which is managed by r* .

1. Is the schema $(R, \mathcal{F})$ in BCNF? If not, why?
2. Are the following decompositions of R in BCNF?
 - (a) $R_1(E, D)$ and $R_2(D, R)$
 - (b) $R_3(E, D)$ and $R_4(E, P, R)$
 - (c) $R_5(D, R)$ and $R_6(D, E, P)$
 - (d) $R_7(D, R)$ and $R_8(E, D)$ et $R_9(E, P)$
3. Use the decomposition algorithm to generate a BCNF decomposition.

Exercise 5

Consider the schema $R(A, B, C, D)$ with $\mathcal{F} = \{AB \rightarrow C, C \rightarrow D, D \rightarrow A, C \rightarrow A, AB \rightarrow D\}$.

1. What are the keys of the relation?
2. Check whether the schema is 3NF or BCNF. Explain why.
3. If necessary, give the BCNF decomposition.